IY453 Stage 1

Assessment Details	Please Complete All Details
Group	B.
Module Title	IY453 Software Design and Implementation.
Assessment Type	Coursework.
Module Tutor Name	Shore Jonathan.
Student ID Number	T0486024
Date of Submission	30/1/2026
Word Count	

- [x] *I confirm that this assignment is my own work. Where I have referred to academic sources, I have provided in-text citations and included the sources in the final reference list.*

- [x] *Where I have used AI, I have cited and referenced appropriately.

The program purpose

The program's purpose is to design a text-based adventure game that allows players to have more interaction by relocating them to a fictional world to help them solve problems and make decisions. This game relies on descriptive writing and user input.

This game has functional and non-functional requirements. The functional requirements are like player interaction, engagement, and error handling. The non-functional requirements of the game are like code quality, user experience, testing, debugging, and documentation.

The game uses at least 30 binary choices in each scenario to meet the functional requirements. Some choices lead to a "Game Over" state, while others allow progress. Additionally, the game has to provide clear feedback for its choices.

The IPO (input, process, and output table)

Feature / Task	Input	Process	Output
Start game	Player name (string)	Store name in Player object	Personalized welcome message
Main menu selection	Menu choice (New Game / Load Game)	Validate choice and branch program flow	Game starts or saved game loads
Scene navigation	Binary choice (1 or 2)	Validate input and determine next scene	New scenario description
Risk-based decision	Binary choice	Apply consequences based on choice	Health change, item gain/loss, or Game Over
Puzzle challenge	Puzzle answer (string or number)	Compare input with correct solution	Puzzle success or failure message
Puzzle reward	Correct puzzle answer	Increase score and unlock next scene	Score update and progression
Puzzle penalty	Incorrect puzzle answer	Reduce health or end game	Damage message or Game Over
Combat initiation	Fight / Run choice	Determine combat or escape logic	Combat start or retreat message
Combat resolution	Random number (system)	Add random value to combat ratings	Win or lose combat outcome
Enemy defeated	Combat win	Update score and possibly drop item	Victory message and score increase
Item discovery	Pick up item (Yes / No)	Add item to inventory if space allows	Item added confirmation
Inventory viewing	Inventory command	Retrieve stored inventory items	Display list of collected items
Item usage	Selected inventory item	Modify player stats (health, attack, defense)	Updated stats displayed
Food or potion use	Item selection	Restore health or boost stats	Health restored message

Feature / Task	Input	Process	Output
Equipment handling	Weapon or armor selection	Apply stat bonuses	Increased attack or defense
Locked path access	Use key item	Check inventory for key	Door opens or access denied
Save game	Save command	Serialize player data to file	Save success or error message
Save file naming	File name (text)	Create or overwrite save file	Confirmation message
Load game	Load command	Read game state from file	Game state restored
Player defeat	Health reaches zero	Trigger game-over logic	Game Over screen
Game completion	Final victory condition	Calculate final score	Victory message and final score

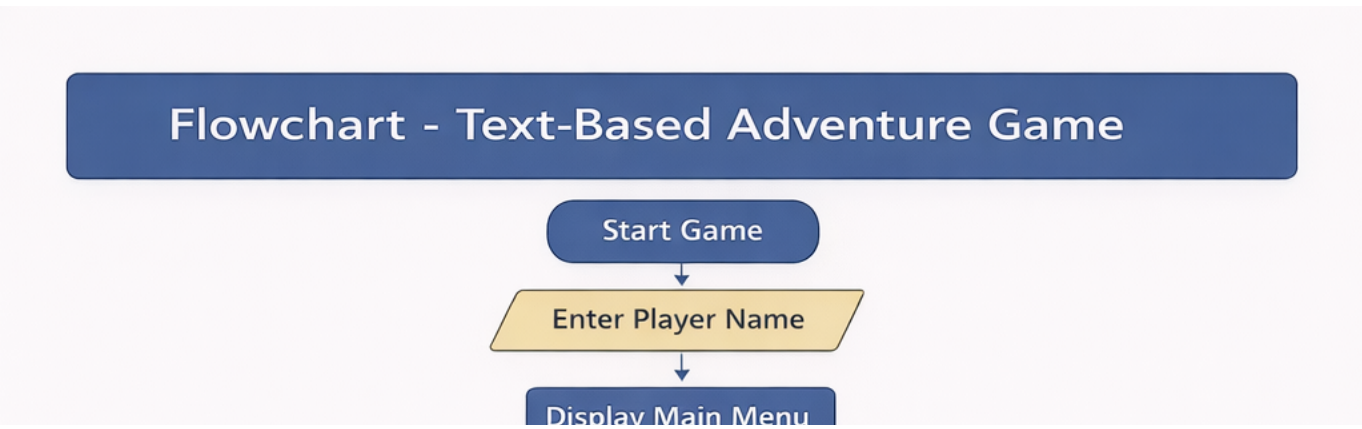
Full reference for this AI tool: ChatGPT. (2026). An IPO table for this text-based adventure game. OpenAI.

prompt: Please give me an IPO table for this text-based adventure game.

Algorithm design

Flowchart:

(ChatGPT, 2026)



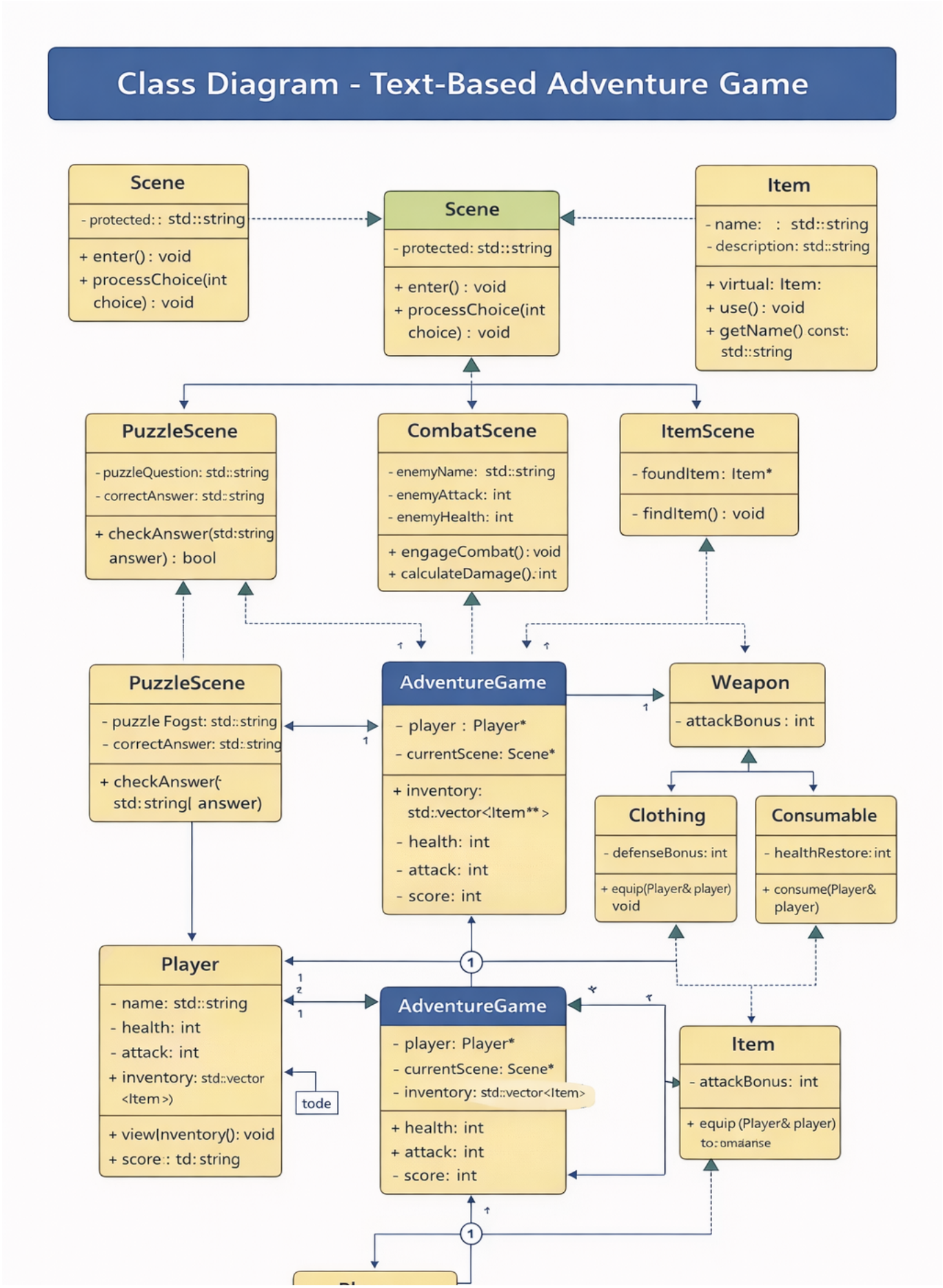


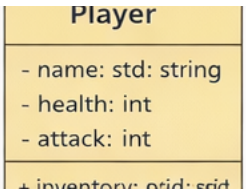
Full reference for this AI tool: ChatGPT. (2026). A flowchart for this text-based adventure game. OpenAI.

prompt: Please give me a flowchart for this text-based adventure game.

Class diagram:

(ChatGPT, 2026)





Full reference for this AI tool: ChatGPT. (2026). A class diagrasm for this text-based adventure game. OpenAI.

prompt: Please give me a class diagram for this text-based adventure game.